# ROSSMANN STORE SALES - SALES PREDICTION REPORT

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             SALES PREDICTION REPORT
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1. PROJECT OVERVIEW
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Project: Rossmann Store Sales Prediction
Task: Task 3 - Sales Prediction

Objective:
To predict daily store sales using Machine Learning
models based on historical sales and store information.


2. DATASET INFORMATION
------------------------------------------------------------
Training Records: 1,017,209
Testing Records: 41,088
Store Records: 1,115

Training Features: 27

Training Date Range:
2013-01-01 00:00:00 to 2015-07-31 00:00:00


3. DATA PREPROCESSING
------------------------------------------------------------
- Cleaned datasets were used.
- Training and testing datasets were merged with store data.
- Missing values were checked.
- Categorical variables were encoded.
- Numerical features were retained.


4. FEATURE ENGINEERING
------------------------------------------------------------
- Year
- Month
- Day
- Day of Year
- Week of Year
- Weekend Indicator
- Month Start and Month End
- Competition Age
- Promotion Activity


5. MODEL DEVELOPMENT
------------------------------------------------------------
The following Machine Learning models were developed:

1. Linear Regression
2. Random Forest Regressor
3. Tuned Random Forest Regressor


6. MODEL EVALUATION
------------------------------------------------------------
Evaluation Metrics:
- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- R2 Score

MODEL COMPARISON:

              Model    RMSE     MAE   R2
Tuned Random Forest 1177.08  794.05 0.85
      Random Forest 1363.03  928.15 0.80
  Linear Regression 2664.04 1946.24 0.24


7. BEST MODEL
------------------------------------------------------------
Selected Model: Tuned Random Forest
RMSE: 1177.08
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MAE: 794.05
R2 Score: 0.8516

8. HYPERPARAMETER TUNING
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RandomizedSearchCV was used for Random Forest
hyperparameter optimization.

Best Parameters:
{'model__n_estimators': 80, 'model__min_samples_split': 5, 'model__min_samples_leaf': 2, 'model__max_featur

Best Cross-Validation RMSE:
2175.26

9. FEATURE IMPORTANCE
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Top 15 Features:

| Feature | Importance |
|---|---|
| numerical__competitiondistance | 0.18 |
| numerical__store | 0.15 |
| numerical__promo | 0.14 |
| numerical__competitionopensinceyear | 0.06 |
| numerical__competitionopensincemonth | 0.06 |
| numerical__dayofweek | 0.06 |
| numerical__competition_age_months | 0.05 |
| numerical__dayofyear | 0.04 |
| numerical__promo2sinceyear | 0.03 |
| numerical__day | 0.03 |
| numerical__promo2sinceweek | 0.03 |
| categorical__storetype_b | 0.03 |
| numerical__weekofyear | 0.02 |
| categorical__assortment_c | 0.01 |
| categorical__storetype_a | 0.01 |

10. VALIDATION PREDICTIONS
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Total Validation Predictions:
175,084

Actual and predicted sales were compared
to evaluate model performance.

11. FINAL TEST PREDICTIONS
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Total Test Predictions:
41,088

Final sales predictions were generated
for the test dataset.

12. OUTPUT FILES
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MODELS:
- linear_regression_model.pkl
- random_forest_model.pkl
- tuned_random_forest_model.pkl
- best_sales_prediction_model.pkl

PREDICTIONS:
- model_comparison.csv
- validation_predictions.csv
- final_sales_predictions.csv
- feature_importance.csv

VISUALIZATIONS:
- 01_model_comparison.png
- 02_actual_vs_predicted.png
- 03_residual_distribution.png
- 04_feature_importance.png

REPORTS:
- Sales_Prediction_Report.txt
- Sales_Prediction_Report.pdf

13. CONCLUSION
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Three Machine Learning models were developed and

evaluated for Rossmann store sales prediction.

The best-performing model was selected based on
validation performance.

Feature importance analysis was performed to
identify important sales prediction factors.

Final predictions were generated for the test dataset.

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END OF REPORT
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